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General Knowledge

Indian Economy

STUDY NOTES

Food Security

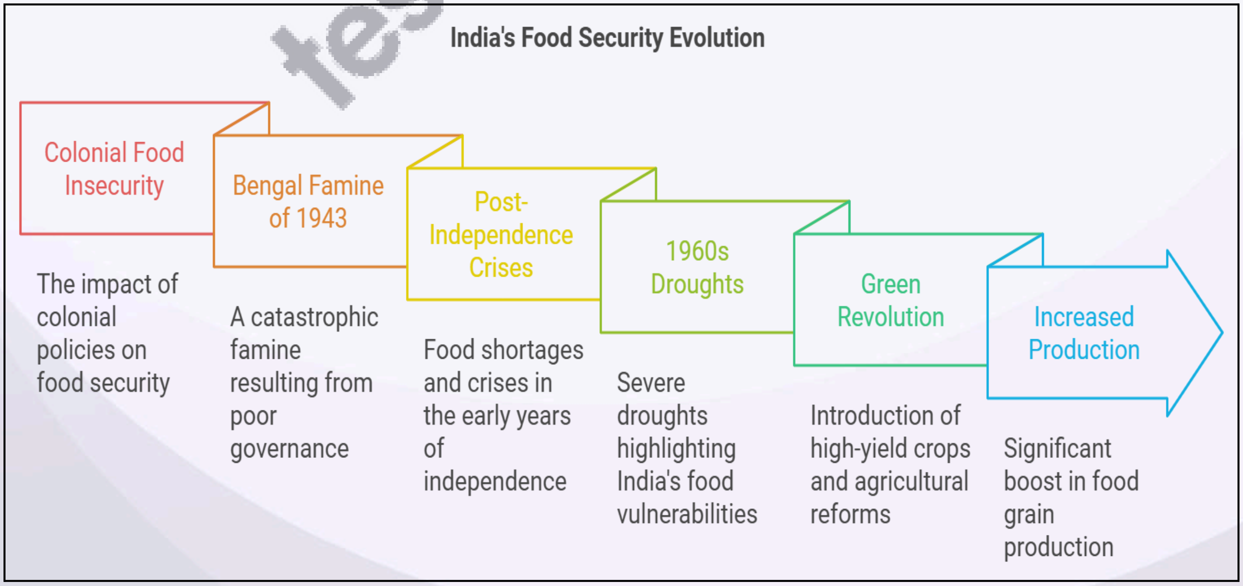
FOOD SECURITY

Introduction

- Food security ensures that all individuals, at any time, have reliable access to sufficient, nutritious, and safe food to lead active and healthy lives.
- As per the Food and Agriculture Organization (FAO), food security has four main dimensions: availability, accessibility, utilization, and stability.
- Achieving food security is crucial to reducing hunger and malnutrition, promoting economic stability, and fostering social harmony.

Historical Background

- India's food security journey is deeply rooted in its colonial history, particularly highlighted by the Bengal Famine of 1943, which claimed 2-3 million lives due to starvation and poor governance.
- Post-independence, India faced multiple food crises, especially during the droughts of the 1960s, which exposed India's vulnerability and reliance on food imports.
- The Green Revolution in the late 1960s marked a turning point, boosting food grain production but also leading to ecological concerns due to the intensive use of chemical inputs and water.



Current Scenario

Self-sufficiency in Staple Grains

- The Green Revolution (focused on grains like **rice and wheat**) and the **White Revolution** (milk production), India has become largely self-sufficient in producing staple foods.
- Self-sufficiency is primarily limited to rice and wheat, while the production of pulses, oilseeds, and other essential crops still lags, necessitating imports to meet demand.
- Certain regions, especially the north-western states like Punjab and Haryana, produce more food, whereas other regions rely heavily on imports due to lower agricultural productivity.

Equitable Access and Distribution Challenges

- Public Distribution System (PDS) Gaps: Although PDS aims to ensure food access for low-income groups, inefficiencies, corruption, and exclusion errors impact its reach and reliability.
- Migrant Population Vulnerability: Migrant workers and urban poor, especially in unorganized sectors, often lack access to subsidized food due to rigid address-based ration card requirements.

- One Nation, One Ration Card (ONORC): This initiative aims to address these issues by making ration cards portable, thus benefiting migrant workers and enhancing food security for transient populations.

Dietary Diversity and Nutritional Security

- Shift from Cereals to High-Value Foods: Rising incomes have led to increased consumption of high-value foods (e.g., fruits, vegetables, dairy, eggs), yet cereals dominate rural diets due to affordability.
- High Dependence on Carbohydrate-Rich Diets: For a large segment of the population, diets lack protein, vitamins, and minerals, leading to widespread micronutrient deficiencies, including Vitamin A, iron, and iodine.
- Promoting Nutritious Crops: Government initiatives are encouraging the cultivation and consumption of pulses, millets, and other nutrient-dense foods to improve dietary diversity.

Malnutrition and Hunger

- Chronic Undernutrition: India has one of the highest rates of undernourished people, with 195 million individuals affected.
- This situation contributes to India's low ranking on the Global Hunger Index.
- High Levels of Child Malnutrition:
  - Stunting (low height for age) affects 38.4% of children under five, indicating chronic undernutrition that impedes growth and cognitive development.
  - Wasting (low weight for height), observed in 21% of children under five, is often linked to acute malnutrition and has severe health implications.
- **Anemia**: A staggering 51.4% of women of reproductive age are anemic, affecting maternal and child health.
- Impact on Human Capital: Persistent malnutrition among children has long-term consequences, reducing educational attainment, productivity, and economic potential.

Impact of Climate Change and Environmental Factors

- Vulnerability to Extreme Weather: With climate change causing irregular rainfall patterns, droughts, and floods, agricultural productivity and food security face increasing risks.

- Water Scarcity: The heavy reliance on water-intensive crops, especially in regions like Punjab, depletes groundwater levels, threatening long-term food sustainability.
- Soil Degradation: Overuse of fertilizers and pesticides, especially in Green Revolution areas, has led to soil degradation, affecting crop yields and reducing food availability.

Economic Disparities and Poverty

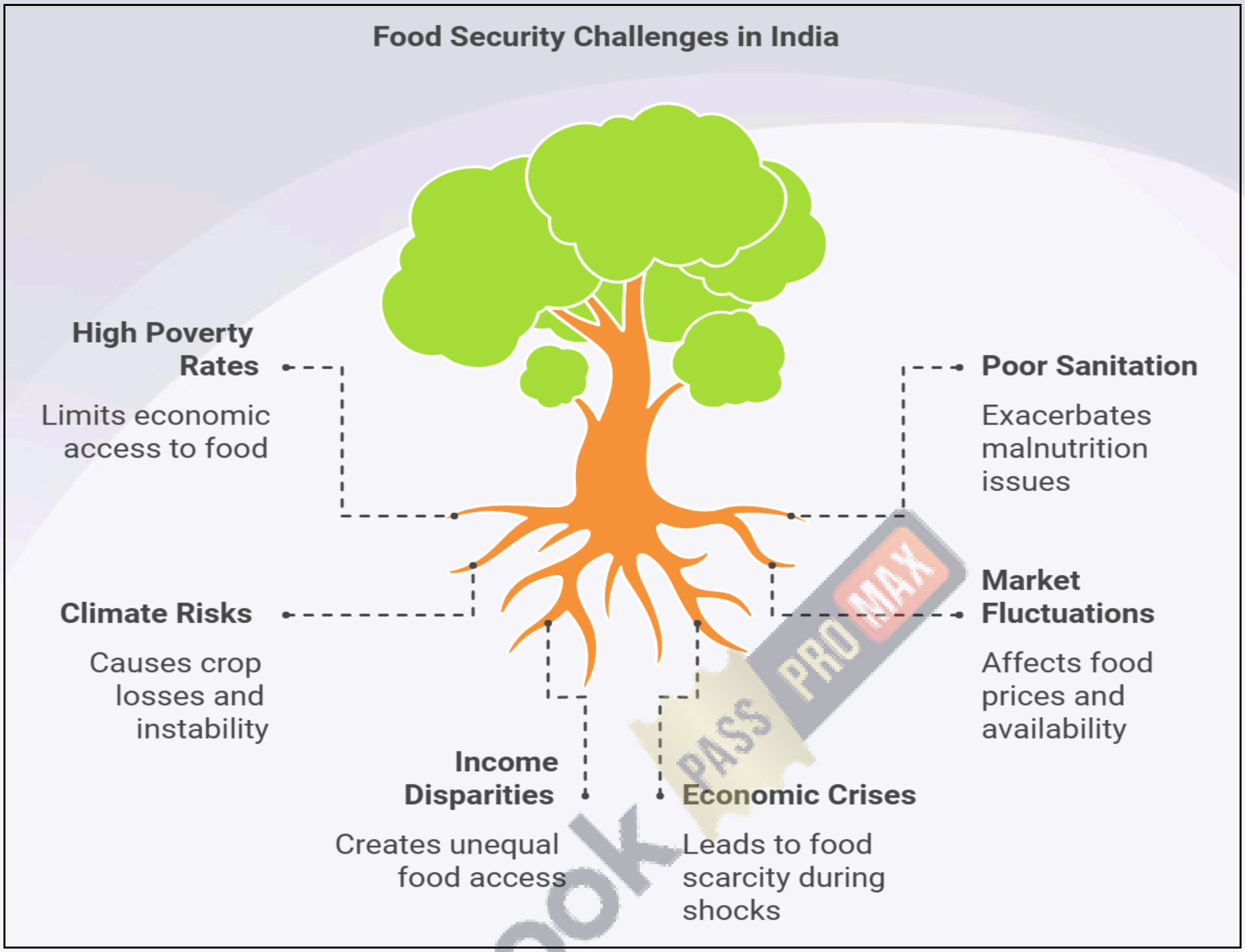
- Income Inequality: Large segments of the population, particularly in rural areas, live below the poverty line, limiting their purchasing power for nutritious foods.
- Rural-Urban Divide: Urban areas generally have better access to diverse foods, while rural regions face challenges in accessing a variety of nutritious foods.

Food Price Volatility

- Impact of Inflation on Food Accessibility: Rising food prices, driven by inflation and supply chain disruptions, particularly impact low-income households, reducing their ability to purchase nutritious foods.
- Essential items like pulses and edible oils often see price fluctuations, creating affordability challenges.
- India imports certain food items, such as edible oils and pulses. Disruptions in global supply chains or international market fluctuations directly affect domestic food prices, influencing food security.

Key Issue	Description	Impact
Agricultural Productivity	Despite high production, yields per hectare for crops like rice and wheat in India are lower than the global average (e.g., 2,992 kg/ha in India vs. over 7,000 kg/ha in North America).	Low productivity strains the food supply and limits India's ability to meet domestic demand efficiently.
	The Green Revolution's focus on rice and wheat led to the neglect of other nutritious crops, affecting crop diversity and degrading soil health.	Limited crop diversity and soil depletion, especially in regions like Punjab and Haryana, reduce the sustainability and resilience of food systems.

Nutritional Quality and Food Diversity	Economic growth has shifted diets from cereals to high-value foods (fruits, vegetables, animal products), yet a balanced diet is still inaccessible to many.	Poor nutrition, as reflected by high iron-deficiency anemia (51.4% among women of reproductive age), affects maternal and child health and productivity.
Public Distribution System (PDS) Challenges	PDS provides a food safety net, but issues like leakage, corruption, and poor targeting reduce effectiveness.	Many needy individuals are excluded due to poverty line inconsistencies, impacting equitable food access.
	The One Nation, One Ration Card initiative seeks to improve PDS effectiveness, especially for migrant workers.	Increases access to PDS benefits for migrant populations, potentially reducing urban food insecurity.
Climate Change and Resource Degradation	Erratic weather, rising temperatures, and extreme events threaten agriculture. Heavy reliance on water-intensive crops (like paddy and sugarcane) worsens water scarcity.	Increased vulnerability of food supply to climate impacts; depletion of water resources stresses agriculture and impacts long-term food stability.
Overpopulation and Urbanization	Rapid population growth and rural-urban migration contribute to overcrowded slums with limited access to nutritious, affordable food.	Increased food insecurity and malnutrition among urban poor populations, as slum areas often lack the infrastructure to support food access and nutrition programs.



Food Security: An Overview

Four Dimensions of Food Security

Availability

- Food availability depends on agricultural production, imports, stock levels, and distribution.
- India is among the world's largest producers of wheat, rice, and pulses, contributing to food availability domestically.
- In 2022, India produced over 120 million tons of rice, marking it as one of the world's largest rice producers.

Accessibility

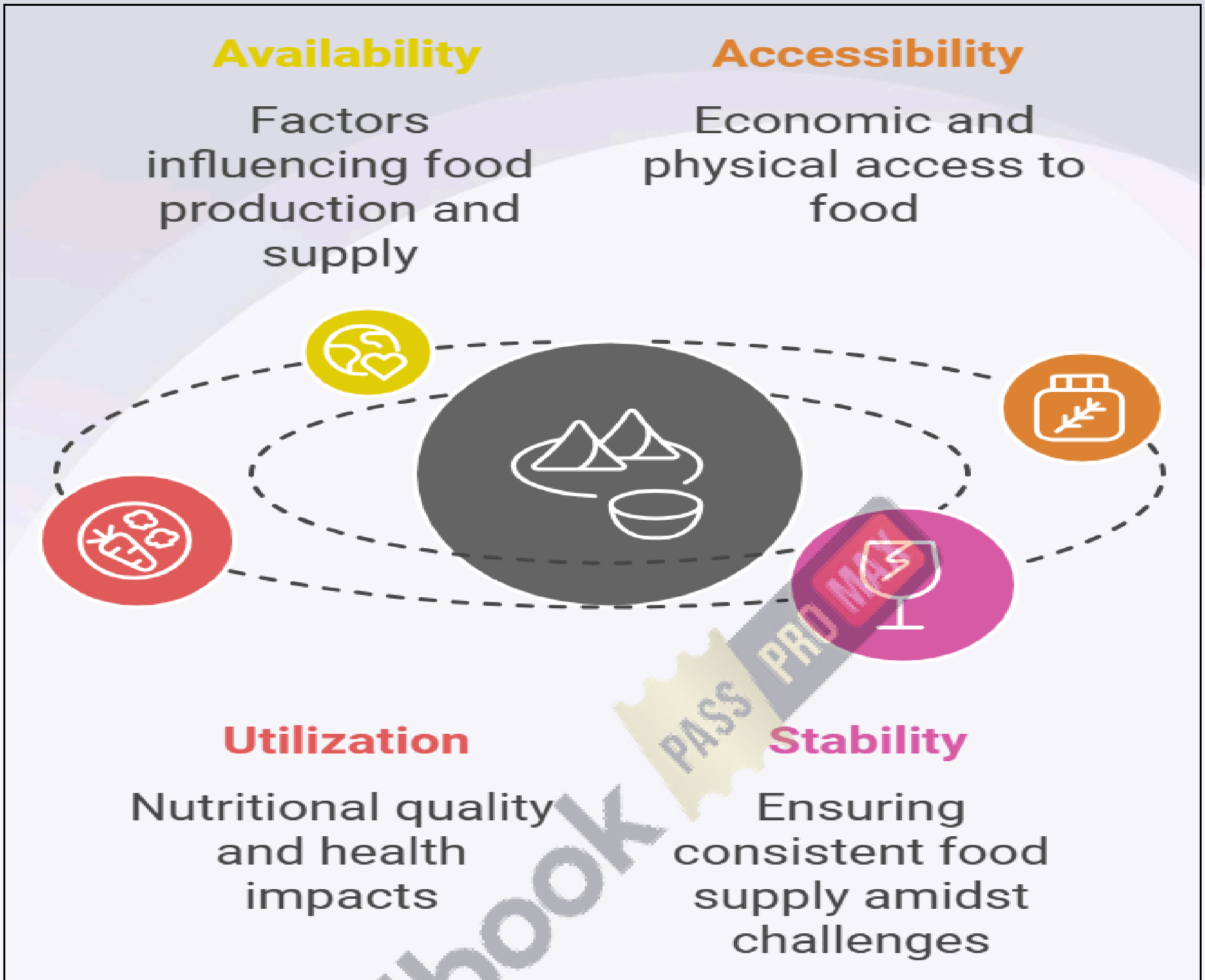
- Food accessibility emphasizes economic and physical access to food, influenced by market dynamics, poverty levels, and food prices.
- Accessibility is often hindered by high poverty rates and disparities in income, especially in rural regions.
- According to NITI Aayog's Multidimensional Poverty Index (2021), over 25% of India's population is multidimensionally poor, impacting food accessibility.

Utilization

- Food utilization refers to the body's ability to consume and effectively metabolize food nutrients, relying on food quality, health, and sanitation.
- Malnutrition, especially in children, is a major issue despite the availability of food. Poor sanitation exacerbates undernutrition.
- India ranks 107th out of 121 countries on the Global Hunger Index 2022, reflecting challenges in food utilization.

Stability

- Stability is essential to ensure that individuals do not face food scarcity during economic shocks or natural disasters.
- Climate-related risks, market fluctuations, and economic crises affect food stability in India.
- India has experienced significant crop losses due to climate extremes, with agriculture bearing 70% of economic losses from natural disasters, according to the FAO.



Importance of Food Security in India

Nutritional and Public Health Impact

- Ensuring food security is key to combating malnutrition and related health issues. Malnutrition affects learning, productivity, and growth in children.
- Nearly 35.5% of children in India are stunted, as reported by the National Family Health Survey (NFHS-5, 2019-2021).

Economic Stability and Growth

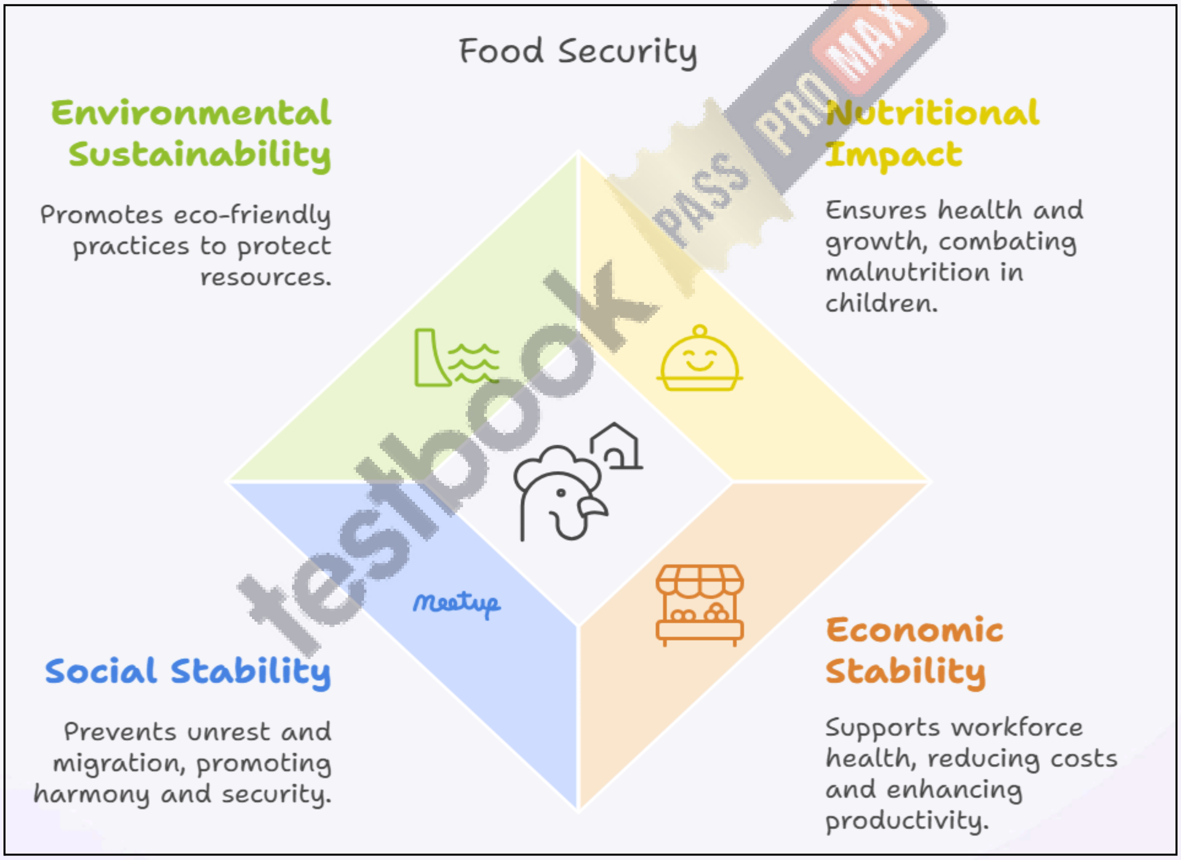
- Food security supports a healthy workforce, reducing healthcare costs and enhancing productivity.
- A World Bank report highlights that India loses around 4% of GDP annually due to malnutrition.

Social Stability and National Security

- Food insecurity can lead to unrest and migration, impacting social harmony and security.
- Addressing food security is also part of India's commitment to the United Nations Sustainable Development Goal (SDG) 2 – Zero Hunger by 2030.

Environmental Sustainability

- Sustainable agricultural practices are vital for ensuring long-term food security without depleting resources.
- India's agriculture sector accounts for about 18% of its greenhouse gas emissions, underscoring the need for eco-friendly farming practices.



Challenges to Food Security in India

High Levels of Poverty

- Economic disparities and lack of income prevent many from accessing adequate food. Poverty exacerbates food insecurity, especially in marginalized communities.

Agricultural Productivity Constraints

- Limited use of modern technology, fragmented landholdings, dependency on monsoons, and inadequate irrigation reduce productivity.
- Nearly 52% of India's cultivable land depends on rainfall, making it vulnerable to droughts.

Malnutrition and Health Challenges

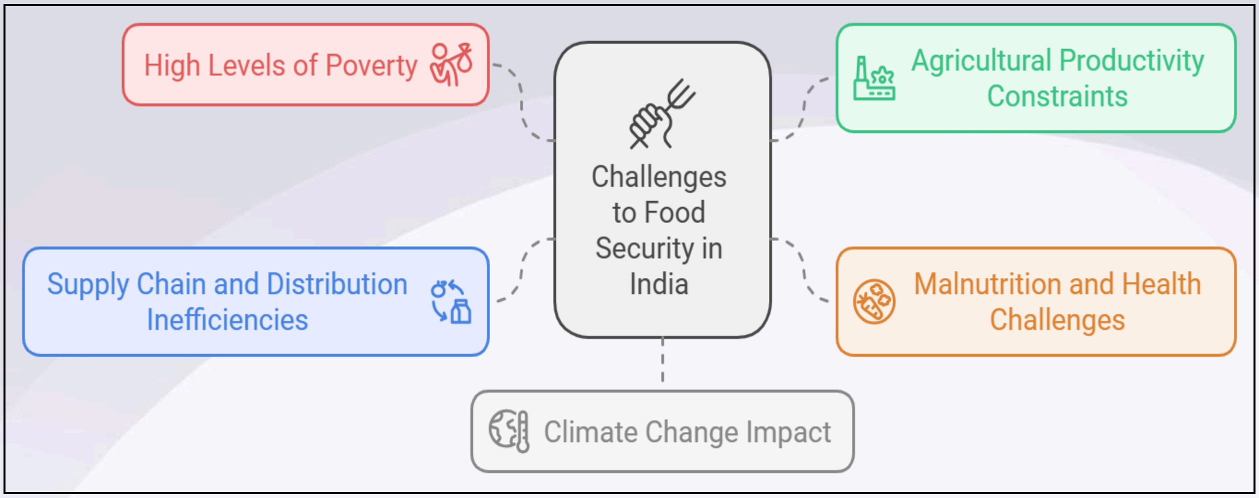
- India faces a triple burden of malnutrition: undernutrition, micronutrient deficiency, and rising obesity.
- According to the FAO, India has one of the highest numbers of undernourished people globally, approximately 195 million.

Supply Chain and Distribution Inefficiencies

- Inadequate storage facilities, cold chains, and poor transportation contribute to post-harvest losses.
- The Central Institute of Post-Harvest Engineering and Technology (CIPHET) estimates India's annual food wastage at around 92,000 crores due to storage and distribution issues.

Climate Change Impact

- Climate-related events like floods, droughts, and changing rainfall patterns significantly impact crop yields.
- According to the Indian Council of Agricultural Research (ICAR), climate change could reduce yields of staple crops by 10-40% by 2030.



Government Initiatives and Policies

National Food Security Act (NFSA), 2013

- Aims to provide subsidized food grains to nearly 67% of the population, focusing on vulnerable groups.
- Includes provisions for Mid-Day Meals, Integrated Child Development Services, and maternity benefits.
- NFSA covers approximately 813 million beneficiaries through the Public Distribution System (PDS).

Public Distribution System (PDS)

- It provides essential food grains like rice and wheat at subsidized rates to BPL families.
- Recent initiatives focus on digitizing the PDS to minimize leakages.
- The "One Nation, One Ration Card" scheme allows migrant workers to access PDS benefits anywhere in India.

Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY)

- It was launched during the COVID-19 pandemic, PMGKAY distributed free food grains to 800 million people affected by lockdowns and economic hardships.

Integrated Child Development Services (ICDS)

- It provides nutrition, preschool education, and health services to children under six, pregnant women, and lactating mothers.
- ICDS reaches nearly 80 million beneficiaries across 1.4 million Anganwadi centres.

Mid-Day Meal Scheme (MDM)

- Offers nutritious meals to school children, improving attendance and nutrition levels.
- The scheme reaches around 120 million children in over 1.2 million schools.

Global Case Studies and Best Practices

Brazil's Zero Hunger Program

- Focuses on food security with policies on social protection, poverty reduction, and rural development.
- India can replicate Brazil's focus on dietary diversity and the inclusion of local foods.

Green Revolution in Africa

- It aims to improve agricultural productivity by introducing advanced technology, crop variety, and irrigation.

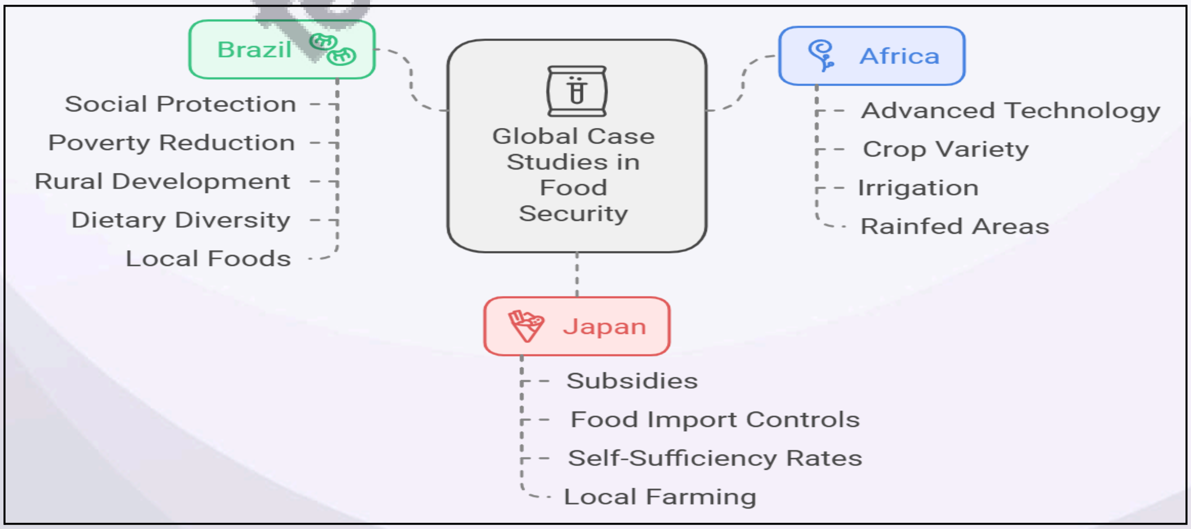
- Lessons can be drawn from India's rainfed areas where technology adoption remains low.

Japan's Self-Sufficiency Policy

- Japan supports domestic agriculture through subsidies and controls food imports to achieve high self-sufficiency rates.
- India can adopt similar policies to strengthen local farming and reduce dependence on imports.

PL-480 Program (Food for Peace)

- The United States PL-480 program (also known as the "Food for Peace" program) was initiated in 1954 to provide food aid to developing countries, including India.
- Due to food shortages and low agricultural productivity in the 1950s and 1960s, India heavily relied on wheat imports from the U.S. under this program.
- The PL-480 imports were critical in stabilizing food supplies in India during times of crisis, especially before the Green Revolution.
- However, this dependence also highlighted India's vulnerability to food scarcity and the need for self-sufficiency, eventually leading to the adoption of policies that fueled the Green Revolution in the late 1960s.
- While the PL-480 program was vital in preventing hunger, it also underscored the importance of agricultural independence, prompting India to improve its agricultural capacity to avoid reliance on foreign aid in the future.



Suggested Measures and Way Forward

Boosting Agricultural Productivity

- Invest in modern technologies such as high-yield seeds, precision farming, and mechanization.
- Support marginal farmers with credit, insurance, and training to adapt to climate-resilient practices.

Enhancing Food Supply Chain Infrastructure

- Develop cold storage facilities, improve transport logistics, and adopt efficient warehousing practices.
- Promote public-private partnerships for better infrastructure and minimize post-harvest losses.

Nutrition-Sensitive Policies

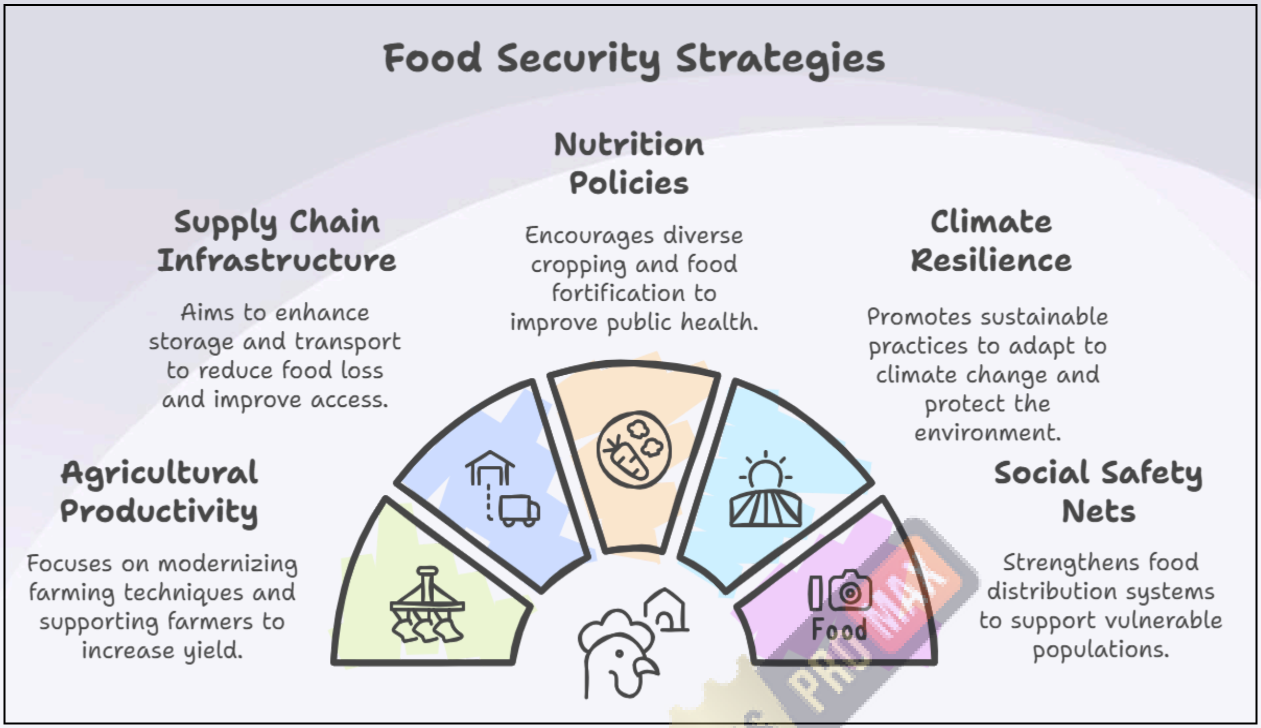
- Diversify cropping patterns to include nutrient-rich crops like pulses, millets, and vegetables.
- Encourage fortification of staple foods and implement programs focusing on maternal and child nutrition.

Climate Resilience and Sustainable Agriculture

- Promote climate-smart agriculture practices, such as water-efficient irrigation and crop diversification.
- The Government's Paramparagat Krishi Vikas Yojana encourages organic farming practices to maintain soil fertility and reduce environmental impact.

Improving Social Safety Nets

- Strengthen PDS, ICDS, and MDM schemes by ensuring effective targeting and minimizing leakage through technology.
- Implement universal food programs in high-poverty areas to cover the most vulnerable populations.


Food Security Strategies
Nutrition Policies
Encourages diverse cropping and food fortification to improve public health.
Supply Chain Infrastructure
Aims to enhance storage and transport to reduce food loss and improve access.
Climate Resilience
Promotes sustainable practices to adapt to climate change and protect the environment.
Agricultural Productivity
Focuses on modernizing farming techniques and supporting farmers to increase yield.
Social Safety Nets
Strengthens food distribution systems to support vulnerable populations.
Food